



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

End Semester Examination - Synoptic

July 2019

Max. Marks: 60.

Duration: 180 Min

Class: S.E.

Semester: IV

Course Code: CE43/IT44

Branch: Computers/IT

Name of the Course: Operating Systems

Instruction:

- (1) All questions are compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Q No.	Question	Max. Mark	CO
Q.1(a)	Differentiate between Symmetric Multiprocessing and Asymmetric Multiprocessing. (Any 6 points) Answer: (1 mark for each correct difference) * 6 - 6 marks	06	CO1
	OR		
Q.1(a)	Differentiate between User mode and Kernel Mode. Answer: (1 Mark for each difference)*6-6 Marks	06	CO1
Q.1(b)	What are the responsibilities of Operating System as Resource Manager? Answer: (1 Mark for each responsibility)*6 - 6 Marks	06	CO1
Q.2	Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4,999. The drive is currently serving a request at cylinder 1,805 and the previous request was at cylinder 2,150. The queue of pending requests, in FIFO order, is: 2020, 1220, 2290, 2810, 540, 350 Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms? a. FCFS b. SSTF c. SCAN d. LOOK Answer: (1 mark for diagram of head movement)*4 - 4 marks (1.5 mark for calculation of individual head movement)*4- 6 marks (0.5 marks for correct total head movement)*4 - 2 marks Note: 0.5 marks will be deducted for each incorrect head movement calculated from entire 3 marks allocated to that algorithm.	12 [3M] [3M] [3M] [3M]	CO2

Q.3(a)	Illustrate and justify the solution for Dining Philosopher problem. (Pseudo-code expected) Answer: Pseudo-code of the solution - 3 Marks Justification of solution - 3 Marks OR Evaluate the following code against all requirements of critical section solution. Consider only 2 processes wants to enter the critical section. Justify each requirement. Code: <pre> do { flag[i] = true; turn = j; while (flag[j] && turn == j); critical section flag[i] = false; remainder section } while (true); </pre> Answer: (2 Mark for justification of each requirement)*3-6 Marks	06	CO3
Q.3(b)	Exemplify the Shortest Job First algorithm for Process scheduling. How exponential averaging burst prediction is used in Shortest Job First algorithm ? Answer: Explanation of SJF - 2 Marks Example of SJF - 2 Marks Exponential Averaging burst prediction - 2 Marks	06	CQ2
Q.4(a)	Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order)? Rank the algorithms in terms of how efficiently they use memory. (diagram is required) Answer: (2 mark for correct answer and analysis using algorithm) * 3 - 6 marks OR	06	CO4

Q.4(a)	Consider a system in which a program can be separated into two parts: code and data. The CPU knows whether it wants an instruction (instruction fetch) or data (data fetch or store). Therefore, two base-limit register pairs are provided: one for instructions and one for data. The instruction base-limit register pair is automatically read-only, so programs can be shared among different users. Analyze the scenario and discuss the advantages and disadvantages of this scheme. Answer: (1 Mark for each advantage)*3-3 Marks (1 Mark for each disadvantage)*3-3 Marks	06	CO4
Q.4(b)	Discuss the different indexing schemes used by Indexed File Allocation method with suitable diagram. Answer: (2 Mark for each scheme)*3 - 6 Marks	06	CO5
Q.5 (a)	Compare between the page replacement algorithms FIFO, Optimal and LRU. Answer: (1 mark for each comparison)*6 - 6 Marks	06	CO4
Q.5(b)	Differentiate between different types of Page Tables. Answer: (1 mark for each difference)*6 - 6 Marks	06	CO4